Method of expressing concentration of solution

21. The molarity of a solution made by mixing 50*ml* of conc. H_2SO_4 (36*N*) with 50 *ml* of water is
(a) 36 *M* (b) 18 *M*
(c) 9 *M* (d) 6 *M*

22. 171 *g* of cane sugar $(C_{12}H_{22}O_{11})$ is dissolved in 1 *litre* of water. The molarity of the solution is
(a) 2.0 M 1.0 *M*
(c) 0.5 *M* d) 0.25 *M*

23. The volumes of $4NHCl$ and $10NHCl$ required to make 1 litre of $6NHCl$ are
(a) 0.75 litre of 10 *N HCl* and 0.25 litre of 4 *N HCl*
(b) 0.25 litre of 4 *N HCl* and 0.75 litre of 10 *N HCl*
(c) 0.67 litre of 4 *N HCl* and 0.33 litre of 10 *N HCl*
(d) 0.80 litre of 4 *N HCl* and 0.20 litre of 10 *N HCl*
(e) 0.50 litre of 4 *N HCl* and 0.50 litre of 10 *N HCl*

24. Which statement is true for solution of 0.020 *M* H_2SO_4
(a) 2 *litre* of the solution contains 0.020 *mole* of SO_4^{2-}
(b) 2 *litre* of the solution contains 0.080 *mole* of H_3O^+
(c) 1 *litre* of the solution contains 0.020 *mole* H_3O^+
(d) None of these

25. 10 *litre* solution of urea contains 240*g* urea. The active mass of urea will be
(a) 0.04 (b) 0.02
(c) 0.4 (d) 0.2

26. 5 *ml* of *N HCl*, 20 *ml* of *N*/2 H_2SO_4 and 30 *ml* of *N*/3 HNO_3 are mixed together and volume made to one litre. The normally of the resulting solution is
(a) $\frac{N}{5}$ (b) $\frac{N}{10}$
(c) $\frac{N}{20}$ (d) $\frac{N}{40}$
(e) $\frac{N}{25}$

27. The amount of $K_2Cr_2O_7$ (*eq.* wt. 49.04) required to prepare 100 *ml* of its 0.05 *N* solution is
(a) 2.9424 *g* (b) 0.4904 *g*
(c) 1.4712 *g* (d) 0.2452 *g*

28. With increase of temperature, which of these changes
(a) Molality
(b) Weight fraction of solute

(c) Fraction of solute present in water

(d) Mole fraction

29. 25*ml* of a solution of barium hydroxide on titration with a 0.1molar solution of hydrochloric acid gave a *litre* value of 35 *ml*. The molarity of barium hydroxide solution was

(a) 0.07 (b) 0.14

(c) 0.28 (d) 0.35

30. 2.0 molar solution is obtained, when 0.5 *mole* solute is dissolved in

(a) 250 *ml* solvent

(b) 250 *g* solvent

(c) 250 *ml* solution

d) 1000 *ml* solvent

31. How many *gram* of *HCl* will be present in $150ml$ of its 0.52 *M* solution

(a) 2.84 *gm* (b) 5.70 *gm*

(c) 8.50 *gm* (d) 3.65 *gm*

32. The number of moles present in 2 *litre* of 0.5 *M* $NaOH$ is

(a) 0.5 (b) 0.1

(c) 1 (d) 2

33. 36*g* water and 828*g* ethyl alcohol form an ideal solution. The mole fraction of water in it, is

(a) 1.0 (b) 0.7

(c) 0.4 (d) 0.1

34. What will be the normality of a solution containing 4.9 *g*. H_3PO_4 dissolved in 500 *ml* water

(a) 0.3 (b) 1.0

(c) 3.0 (d) 0.1

35. 3.0 molal *NaOH* solution has a density of 1.110 *g/ml*. The molarity of the solution is

(a) 3.0504 (b) 3.64

(c) 3.05 (d) 2.9732

36. Which of the following modes of expressing concentration is independent of temperature

(a) Molarity (b) Molality

(c) Formality (d) Normality

37. The molality of a solution is

(a) Number of moles of solute per $1000ml$ of the solvent

(b) Number of moles of solute per $1000gm$of the solvent

(c) Number of moles of solute per $1000ml$ of the solution

(d) Number of gram equivalents of solute per $1000ml$ of the solution

38. The number of molecules in $16gm$ of methane is

(a) 3.0×10^{23} (b) 6.02×10^{23}

(c) $\frac{16}{6.02} \times 10^{23}$ (d) $\frac{16}{3.0} \times 10^{23}$

39. The number of moles of a solute in its solution is 20 and total number of moles are 80. The mole fraction of solute is

(a) 2.5 (b) 0.25

(c) 1 (d) 0.75

40. The normality of a solution of sodium hydroxide 100 *ml* of which contains 4 grams of $NaOH$ is

(a) 0.1 (b) 40

(c) 1.0 (d) 0.4

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